Probability Scales

Name: ________________________ Date: _____________ Year 6 Maths — Probability

ACARA v9.0: AC9M6P01

Learning Intention

I am learning that probabilities lie on numerical scales of 0-1 or 0%-100%, and to assign probabilities using fractions, decimals and percentages.

Success Criteria

- I can place events on a 0 to 1 probability scale.
- I can convert between fractions, decimals and percentages for probability.
- I can assign a probability to a real-world event.

0 Is Impossible, 1 Is Certain

Probability scales run from 0 (impossible) to 1 (certain), or 0% to 100%. A probability of 0.5 or 50% or 1/2 means an event is just as likely to happen as not.

Worked Example

A bag has 4 red and 4 blue balls.

P(red) = 4/8 = 1/2 = 0.5 = 50%

Part A — Warm Up

Circle the probability that means 'certain': **0 0.5 1**

True or False: A probability can be greater than 1. ______

Convert 1/4 to a percentage. ______

Part B — Core Skills

A spinner has 5 equal sections: 2 red, 3 blue. Find P(red) as a fraction. ______

Convert 0.75 to a percentage and a fraction. Percentage: ______ Fraction: ______

Order these probabilities from least to most likely: 0.2, 0.9, 0.5, 0.1

____

Give a real-world event with a probability close to 0 (very unlikely).

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Word Bank

probability • impossible • certain • likely • unlikely • fraction • decimal • percentage • scale

Working Space — Draw a Probability Scale from 0 to 1 and Mark 4 Events on It

Probability Scales — Page 2

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Part C — Challenge: The School Raffle

> *A Geelong school raffle sells 200 tickets. Priya buys 8 tickets. Only 1 winning ticket will be drawn.*

Q8. Calculate Priya's probability of winning as a fraction, then convert it to a percentage (round to 1 decimal place).

Q9. The school decides to sell 100 more tickets (300 total), and Priya buys no more tickets. Explain what happens to her probability of winning and why.

Q10. A student says: "Since 8 out of 200 is a small number, Priya has basically no chance of winning." Do you agree or disagree? Use the probability scale (0 to 1) to explain your reasoning.

Answer Key — Part A

A1: 1

A2: False — probability is always between 0 and 1 (or 0% and 100%).

A3: 25%.

Answer Key — Part B

B1: 2/5.

B2: 75%; 3/4.

B3: 0.1, 0.2, 0.5, 0.9.

B4: Any reasonable very unlikely event (e.g. "it will snow in Geelong in summer") — teacher to check.

Answer Key — Part C

C8: 8/200 = 1/25 = 4.0%.

C9: Her probability decreases, because the total number of tickets increases (to 300) while her number of tickets stays the same (8), making her chance of winning 8/300 instead of 8/200, which is a smaller fraction.

C10: Disagree (or discuss both) — while 8/200 (0.04) is much closer to 0 (impossible) than 1 (certain), it is still a real, non-zero chance, not 'basically no chance'; on the 0–1 scale it sits closer to the impossible end but is not impossible.

I Can Checklist

☐ I can place events on a probability scale.

☐ I can assign probabilities to real events.

☐ I can convert fractions, decimals and percentages.